

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

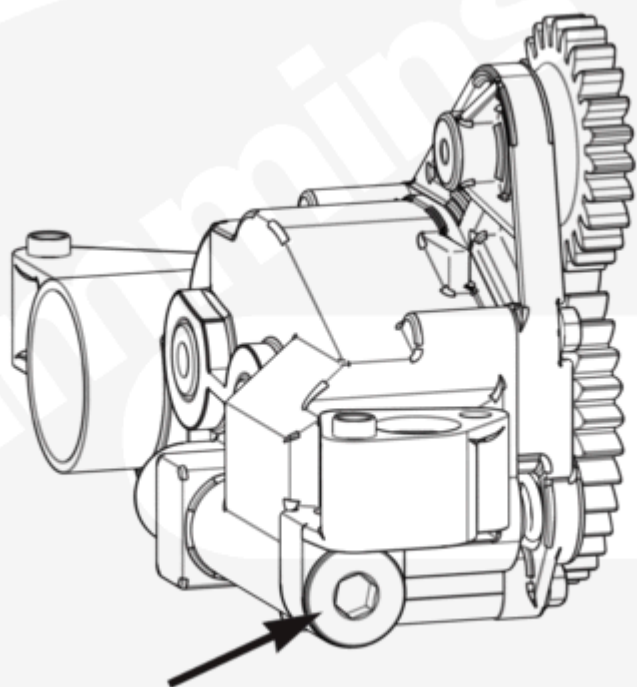
- Spring compression tester, Part Number 5299714, or equivalent
- Spring compression tester, Part Number 3375182.

Additional Service Items

- No additional service items required.

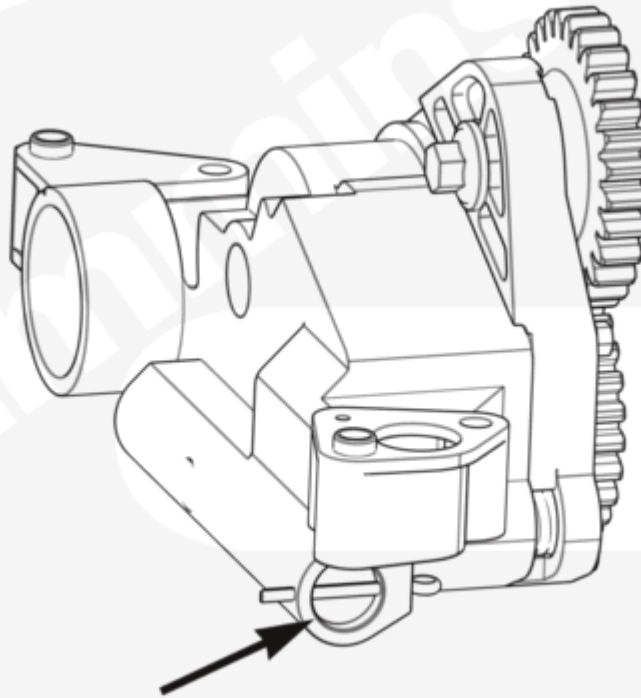
General Information

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07y00021

Threaded Regulator Plug



07y00022

Pinned Regulator Plug

Note : There are two compatible lube pumps in service. One with a pinned regulator plug, and one with a threaded regulator plug. The plug can be found on the bottom of the lubricating oil pump (see above).

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

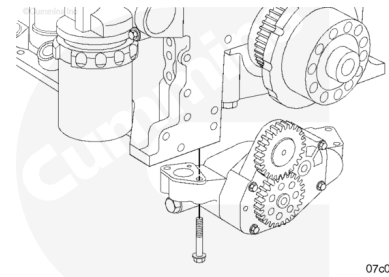
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html)

Remove

⚠ CAUTION ⚠

Do not discard shims, if equipped. They must be installed in the original location. Failure to do so can result in engine damage.



Remove four capscrews, lubricating oil pump, and shims, if equipped.

When installing a new lubricating oil pump, new shims will possibly be needed to acquire acceptable lash.

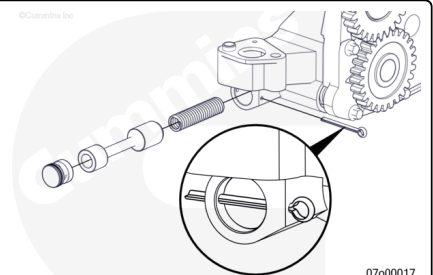
Note : If the lubricating oil pump is found to have generated debris or debris damage is identified in the lubricating oil pump, the lubricating oil filter **must** be examined. If metallic debris is found in the filter, the lubricating oil cooler element **must** be replaced and progressive damage guidelines followed to complete the repair.

Disassemble

With Pinned Regulator Plug

⚠ WARNING ⚠

The pressure regulator spring is under compression. Slowly remove the cap to reduce the possibility of personal injury.

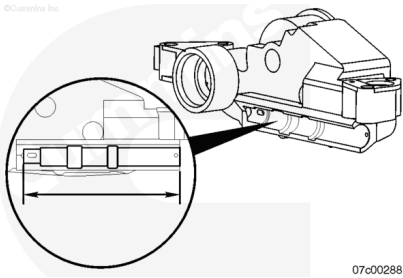


Remove the lubricating oil pressure regulator and spring by removing the split pin that secures the spring cap to the lubricating oil pump.

Discard the o-ring.

Measure the regulator spring bore for beat-in.

Spring Bore Depth Specification		
mm		in
186.95	MIN	7.3602
187.45	MAX	7.380



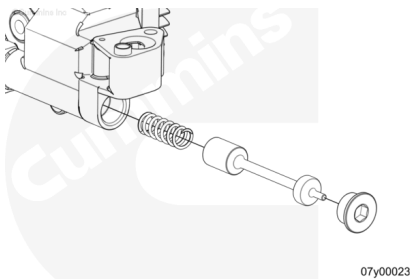
If the regulator spring bore depth is **not** within the specifications, the housing **must** be replaced.

With Threaded Regulator Plug

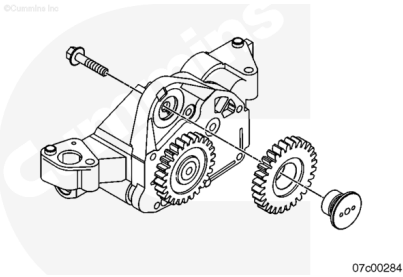
⚠ WARNING ⚠

The pressure regulator spring is under compression. Slowly remove the cap to reduce the possibility of personal injury.

Remove the lubricating oil pressure regulator and spring by removing the threaded plug.



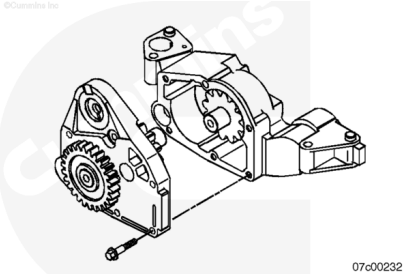
The external idler gear has a left-hand thread capscrew.
Remove the external idler gear from the lubricating oil pump.



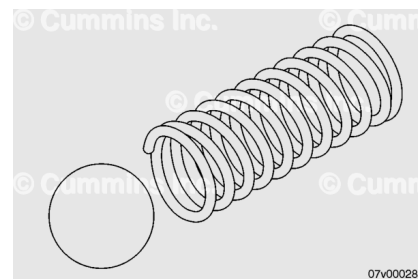
⚠ WARNING ⚠

The high-pressure relief valve is under compression. Apply pressure to the cover while removing the capscrews. Slowly release the pressure to reduce the possibility of personal injury.

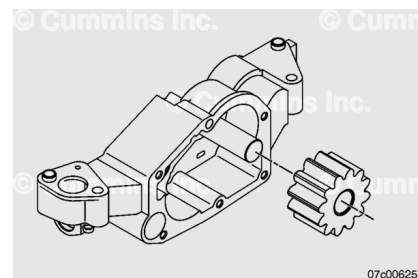
Remove the six assembly capscrews and the lubricating oil pump cover.



Remove the high-pressure relief spring and ball.



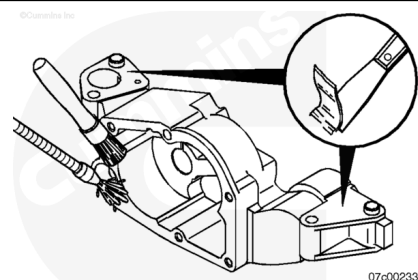
Remove the idler pumping gear from the shaft in the lubricating oil pump body.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

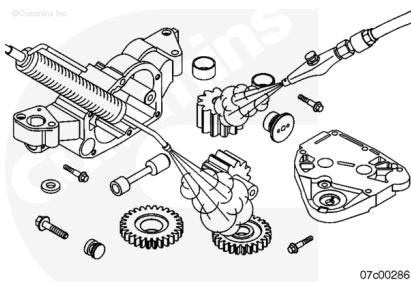
**Do not allow dirt or gasket material to enter the oil passages.
Bearing damage can result.**

Clean the lubricating oil pump components with solvent.

Use a bristle brush to clean any internal ports.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



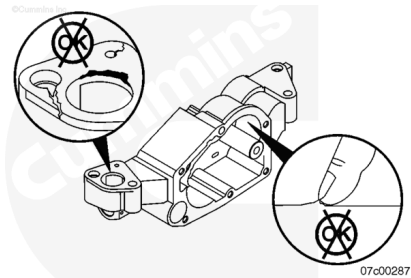
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a steam cleaner to rinse the components.
Dry with compressed air.

Inspect the following areas of the lubricating oil pump for damage or blockage.

- Lubricating oil pump body gear cavity
- Inlet and outlet ports
- Internal oil passages
- Lubricating oil pump housing body
- Pressure regulator bore
- High-pressure relief bore and seat.

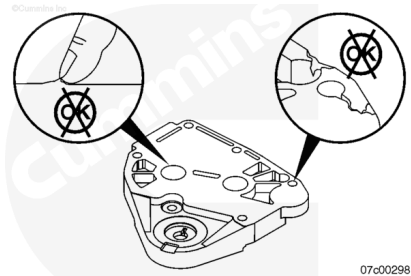


If the lubricating oil pump housing is damaged, it **must** be replaced.

Inspect the lubricating oil pump cover for damage.

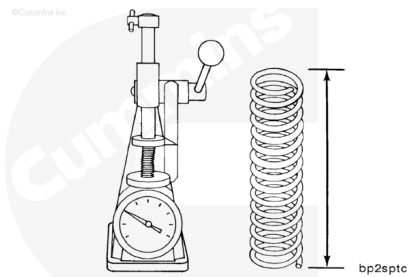
Inspect the bushing for scuffing and wear.

If the marks on the lubricating oil pump cover can be felt with a fingernail, the part **must** be replaced.



With Pinned Regulator Plug

Use a spring compression tester, Part Number 5299714, or equivalent, to measure the pressure regulator valve spring at working height, 70 mm [2.756 in].



Lubricating Oil Pressure Regulator Spring Load at 70 mm [2.756 in]

n.m		ft-lb
131.8	MIN	29.63

Lubricating Oil Pressure Regulator Spring Load at 70 mm [2.756 in]

n.m		ft-lb
135.3	MAX	30.42

If the regulator spring is **not** within the specifications, the spring **must** be replaced.

Use a spring compression tester, Part Number 3375182, to measure the high-pressure relief spring at working height, 45.82 mm [1.804 in].

High-Pressure Relief Valve Spring (Compressed Length) Load at 75.07 mm [2.956 in]

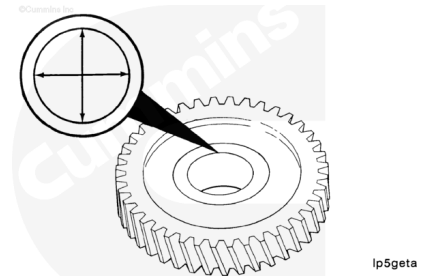
n		ft-lb
281.47	MIN	63.277
298.87	MAX	67.189

If the pressure relief valve spring is **not** within the specification, the spring **must** be replaced.

Measure the inside diameter of the bushing in the external idler gear.

External Idler Gear Bushing Inside Diameter

mm		in
32.044	MIN	1.2616
32.069	MAX	1.2626

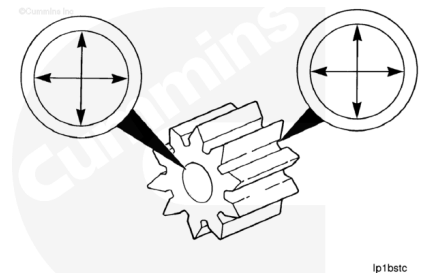


If the external idler gear bushing is **not** within specifications, the bushing **must** be replaced.

Measure the inside diameter of the internal idler gear.

Internal Idler Gear Inside Diameter

mm		in
24.990	MIN	0.9839
25.071	MAX	0.9870



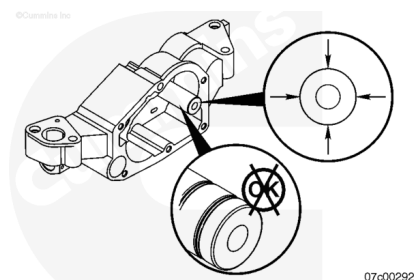
If the internal idler gear is **not** within specifications, it **must** be replaced.

Inspect the internal idler shaft for damage.

Measure the outside diameter of the idler shaft.

Internal Idler Shaft Outside Diameter

mm		in
24.963	MIN	0.9828
24.976	MAX	0.9833



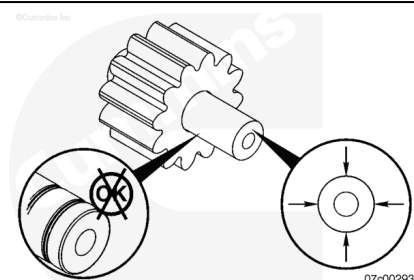
07c00292

Inspect the internal drive gear and shaft assembly for damage.

Measure the outside diameter of the internal drive gear shaft.

Outside Diameter of Internal Drive Gear Shaft

mm		in
24.858	MIN	0.9792
24.871	MAX	0.9787



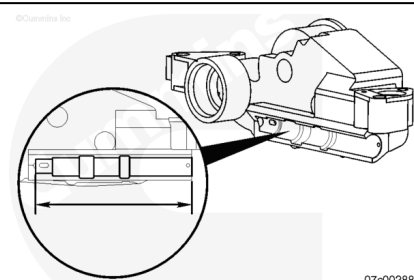
07c00293

With Threaded Regulator Plug

Measure the regulator spring bore for beat-in.

Spring Bore Depth Specification

mm		in
175.3	MIN	6.91
175.6	MAX	6.92



07c00288

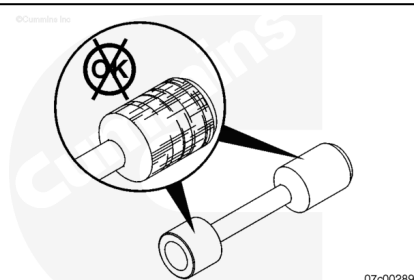
If the regulator spring bore depth is **not** within the specifications, the housing **must** be replaced.

Inspect the pressure regulator spool for scuffing.

Minor spool scuff is acceptable. If the scuff can be felt by a fingernail, the spool will need to be replaced.

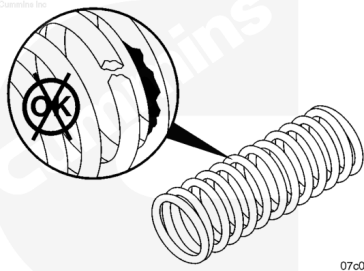
Insert the spool in the bore and check for smooth movement in and out of the bore. If any hesitation is felt, the spool **must** be replaced and the bore inspected for burrs.

Remove any burrs found in the bore before installing a new spool.



07c00289

Inspect the pressure regulator spring for a broken coil or rust.

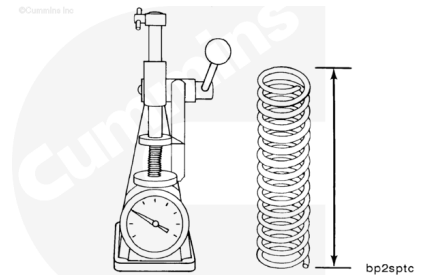


07c00290

Use a spring compression tester, Part Number 5299714, or equivalent, to measure the pressure regulator valve spring at working height, 35.5 mm [1.39 in].

Lubricating Oil Pressure Regulator Spring Load at 35.5 mm [1.39 in]

n		lbf
85.6	MIN	19.3
94.7	MAX	21.3



bp2septc

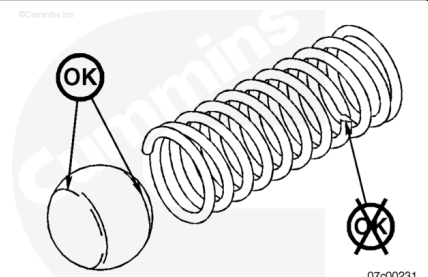
If the regulator spring is **not** within the specifications, the spring **must** be replaced.

Use a spring compression tester, Part Number 3375182, to measure the high-pressure relief spring at working height, 15 mm [0.59 in].

High-Pressure Relief Valve Spring (Compressed Length) Load at 15 mm [0.59 in]

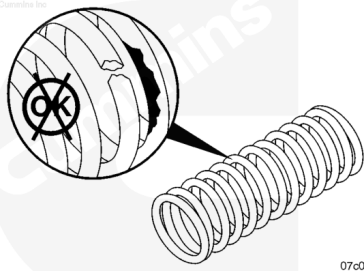
n		lb
366	MIN	82.3
392	MAX	88.1

Inspect the high-pressure relief plunger ball. A slight wear around the plunger from contact with the pump body and spring is acceptable.



07c00231

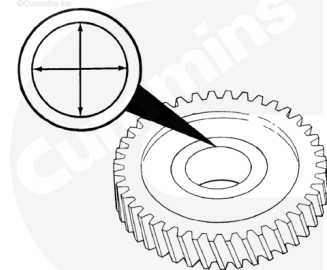
Inspect the high-pressure relief spring for a broken coil or rust.



07c00290

Measure the inside diameter of the bushing in the external idler gear.

External Idler Gear Bushing Inside Diameter		
mm		in
30.04	MIN	1.18
30.06	MAX	1.18



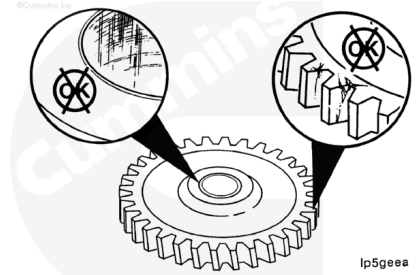
lp5geta

If the external idler gear bushing is **not** within specifications, the bushing **must** be replaced.

Inspect the external drive gear for cracks, chips, and broken teeth.

If the external drive gear is damaged, the gear **must** be replaced.

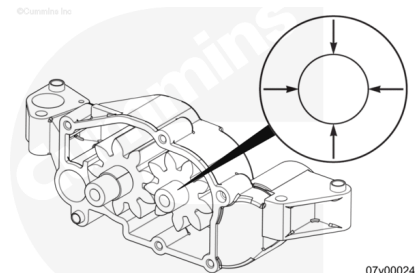
Check the drive gear bushing for wear and scuffing.



lp5geea

Measure the outside diameter of the idler shaft.

Internal Idler Shaft Outside Diameter		
mm		in
24.04	MIN	0.95
24.06	MAX	0.95



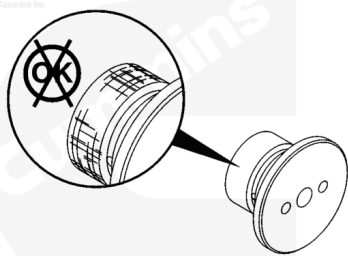
07y00024

If the idler shaft is **not** within specifications, it **must** be replaced.

Discard the mounting bolt.

Inspect the external idler gear retainer.

Minor retainer scuffing is acceptable. If the scuffing can be felt by a fingernail, replace the retainer.

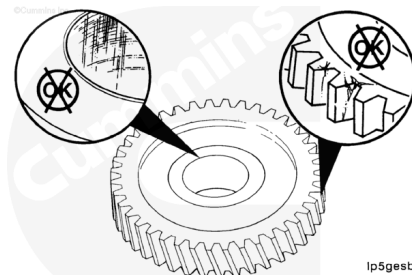


07c00291

Inspect the external idler gear for cracks, chips, and broken teeth.

If the external idler gear is damaged, the housing **must** be replaced.

Check the idler gear bushing for wear and scuffing.

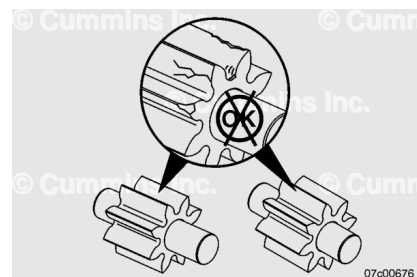


lp5gestb

Inspect the lubricating oil pump internal gears and bushings for excessive wear, cracks, or broken teeth.

Erosion and pitting on the tip and root of the gear tooth is unacceptable. The gear **must** be replaced.

If the drive shaft is **not** within specifications, it **must** be replaced.

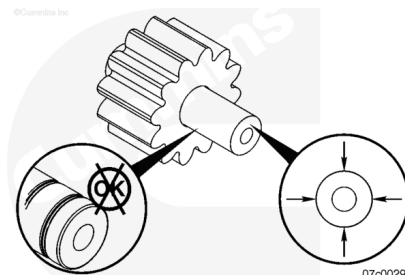


07c00676

Inspect the internal drive gear and shaft assembly for damage.

Measure the outside diameter of the internal drive gear shaft.

Internal Drive Gear Shaft Outside Diameter		
mm		in
24.04	MIN	0.95
24.06	MAX	0.95

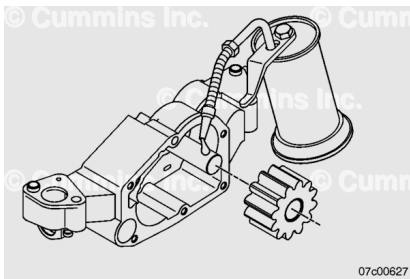


07c00293

Assemble

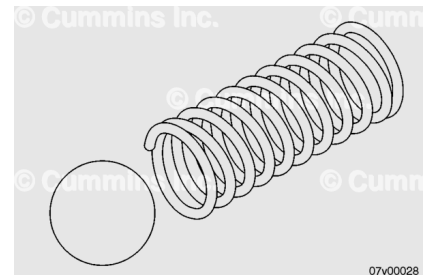
Lubricate the drive shaft with clean engine lubricating oil.

Install the internal idler gear onto the shaft.



Install the ball first, then the spring.

Install the high-pressure relief ball and spring into its bore in the lube pump housing.



With Pinned Regulator Plug

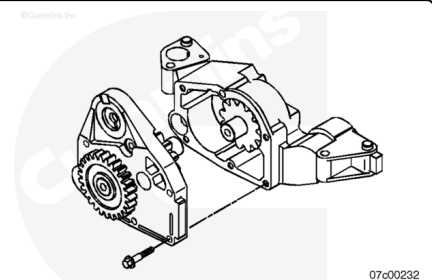
No gasket is used when installing the lube pump cover.

Install the cover to the lube pump.

Tighten the center capscrews first, then the outside capscrews.

Install the capscrews and tighten.

Torque Value: 29 n·m [21 ft-lb]

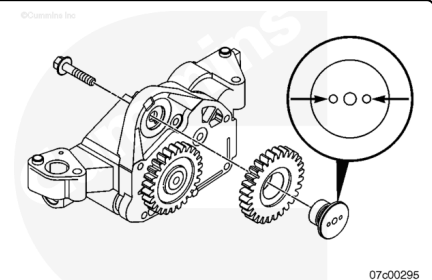


Install the external idler gear.

Insert the external idler gear retainer such that the holes are at the 3 and 9 o'clock positions.

The idler gear retaining capscrew has left-handed threads.

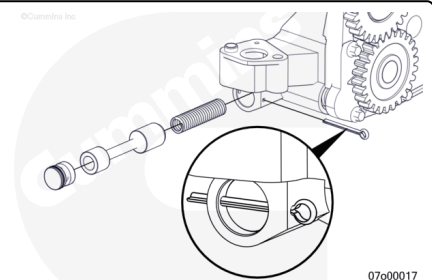
Torque Value: 115 n·m [85 ft-lb]



Lubricate the oil pressure regulator valve and bore with clean engine oil.

Install the spring, valve, and cap with a new o-ring.

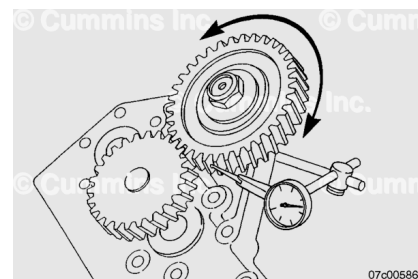
Note : The oil hole of the pressure regulator end cap **must** be installed facing the regulator valve or the lube system will dump the oil back to the pan and will **not** regulate the oil pressure properly. Use a new split pin to secure the cap to the lubricating oil pump body.



Measure the gear backlash by mounting a dial indicator. Position the dial indicator contact point onto the lubricating oil pump external idler gear teeth.

With the external drive gear locked into place and the dial indicator contact in place, rotate the external idler gear **clockwise** and **counterclockwise** to measure the backlash.

Use the backlash measurement and the following table to classify and mark the pump classes.



Lubricating Oil Pump Gear Backlash

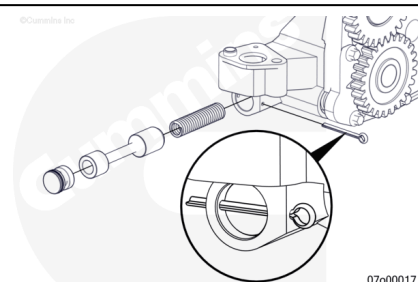
Class	mm	in
1	0.457 to 0.610	0.018 to 0.024
2	0.152 to 0.305	0.006 to 0.012
3	0.000 to 0.152	0.000 to 0.006
4	0.305 to 0.457	0.012 to 0.018

Lubricate the oil pressure regulator valve and bore with clean engine oil.

Install the spring, valve, and cap with a new o-ring.

Note : The oil hole of the pressure regulator end cap **must** be installed facing the regulator valve or the lube system will dump oil back to pan and will not regulate the oil pressure properly.

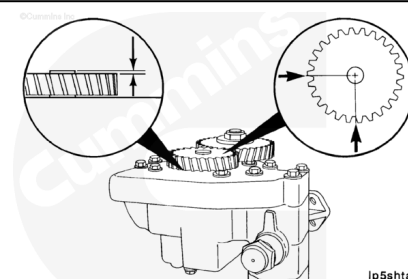
Use a new split pin to secure the cap to the lubricating oil pump body.



Measure the end clearance of the external drive gear.

Drive Shaft End Clearance

mm		in
0.100	MIN	0.003
0.163	MAX	0.006



With Threaded Regulator Plug

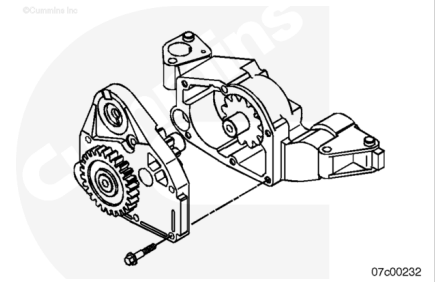
No gasket is used when installing the lube pump cover.

Install the cover to the lube pump.

Tighten the center capscrews first, then the outside capscrews.

Install the capscrews and tighten.

Torque Value: 25 n·m [221 in-lb]

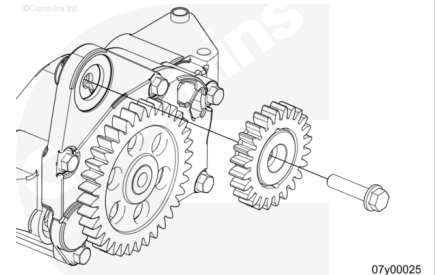


Install the external idler gear.

Insert the external idler gear shaft such that the holes are at the 4 and 11 o'clock positions.

Apply Loctite® 271, Part Number 5394475, to idle gear capscrew

Torque Value: 55 n·m [41 ft-lb]

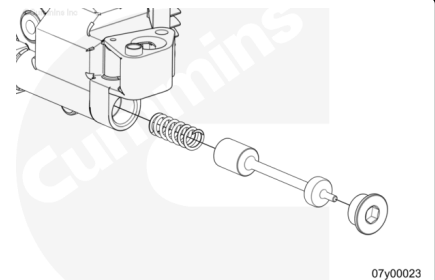


Lubricate the oil pressure regulator valve and bore with clean engine oil.

Install the spring, valve, and threaded plug.

Apply Loctite® 243 Part Number 4919200.

Torque Value: 70 n·m [52 ft-lb]

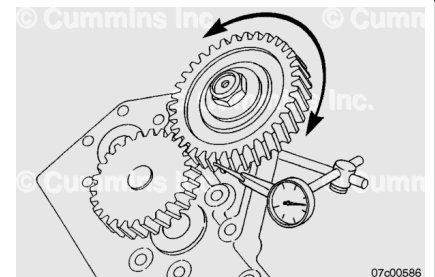


Measure the gear backlash by mounting a dial indicator. Position the dial indicator contact point onto the lubricating oil pump external idler gear teeth.

With the external drive gear locked into place and the dial indicator contact in place, rotate the external idler gear **clockwise** and **counterclockwise** to measure the backlash.

If gear lash is **not** met, replace the entire pump.

Gear Lash Requirement: 0.08 – 0.57 mm [0.003 – 0.022 in].



Install

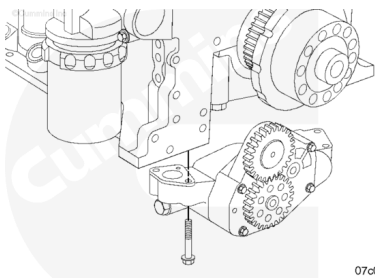
With Pinned Regulator Plug

Align dowels on the lubricating oil pump with mounting holes in the cylinder block.

Install the lubricating oil pump and shims, if they were removed, and the four capscrews.

Tighten the capscrews.

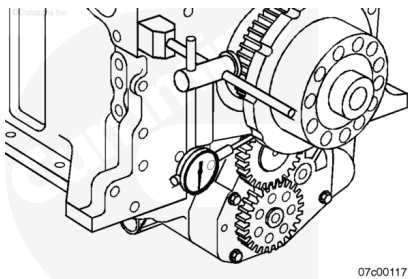
Torque Value: 56 n•m [41 ft-lb]



Use a dial indicator to check the backlash between the lubricating oil pump idler gear and crankshaft gear.

Lubricating Oil Pump with Pinned Regulator Plug Idler Gear Backlash

mm		in
0.18	MIN	0.007
0.32	NOM	0.013
0.46	MAX	0.018

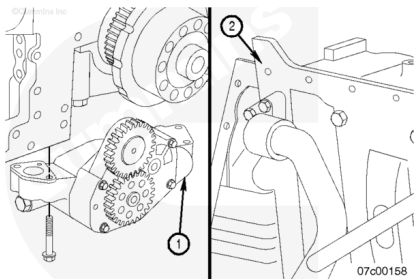


If the backlash is within specification, conduct finishing steps.
If backlash is outside specification, select the proper shims in the next step.

⚠ CAUTION ⚠

Improper gear lash can cause excessive noise, or damage to the gear train and the lubricating oil pump.

Shims will be available in six different thicknesses in order to provide correct lash. The various thicknesses are identified by the color of the shim. If the engine was built prior to December 2014, identify the number stamped on the lubricating oil pump (1) and the letter on the block (2). Use the following table to determine which shim will be used according to the stamped number.



If a lubricating oil pump or cylinder block is replaced, or the engine was built after December 2014, the block and lubricating oil pump will **not** be stamped. In that instance, a shim **must** be chosen and installed. The lubricating oil pump idler gear backlash **must** be checked. If the backlash is **not** within the specification, choose another size shim and recheck the backlash. Do this until the backlash is within specification.

Lubricating Oil Pump	Cylinder Block			
	A	B	C	D

1	0.000 mm [0.000 in] -None-	0.45 mm [0.018 in] -Black-	0.55 mm [0.022 in] -Purple-	0.35 mm [0.014 in] -Yellow-
2	0.45 mm [0.018 in] -Black-	0.65 mm [0.026 in] -Red-	0.75 mm [0.030 in] -Blue-	0.55 mm [0.022 in] -Purple-
3	0.55 mm [0.022 in] -Purple-	0.75 mm [0.030 in] -Blue-	0.85 mm [0.033 in] -Green-	0.65 mm [0.026 in] -Red-
4	0.35 mm [0.014 in] -Yellow-	0.55 mm [0.022 in] -Purple-	0.65 mm [0.026 in] -Red-	0.45 mm [0.018 in] -Black-

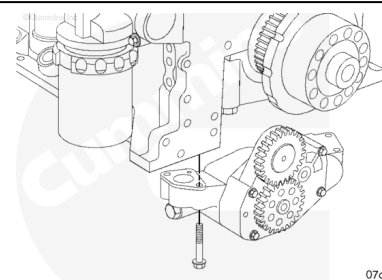
With Threaded Regulator Plug

Align the dowels on the oil pump with the mounting holes in the cylinder block.

Install the oil pump and the four capscrews.

Tighten the capscrews.

Torque Value: 56 n•m [41 ft-lb]

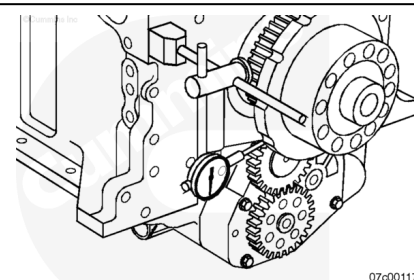


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Use a dial indicator to check the backlash between the lubricating oil pump idler gear and crankshaft gear.

Lubricating Oil Pump with Threaded Regulator Plug
Idler Gear Backlash

mm		in
0.08	MIN	0.003
0.57	MAX	0.022



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If the backlash is within specification, conduct finishing steps.

If backlash is outside specification, replace the entire pump.

Lubricating oil pumps with threaded regulator plugs do **not** use shims to adjust gear lash.

Finishing Steps

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.